Geography GCE A Level 2021 - Paper 3

Duration: 3h15 | Cameroon GCE Board

General Certificate of Education Board - General Certificate of Education Examination. Geography 3 (0750), Paper 3, ADVANCED LEVEL, June 2021. Duration: three and a quarter hours.

Answer FOUR questions in all. Answer question ONE and three others, choosing ONE question from SECTION B and TWO from SECTION C. Question ONE carries thirty-one marks and the rest carry twenty-three marks each.

Only the required number of questions will be marked.

In answering questions on this paper, you are encouraged to refer to your own field and practical work where relevant. Credit will be awarded for such references.

Candidates are advised NOT to attempt the questions on field investigations if such an exercise has not been carried out.

Marks will be awarded for well annotated maps and diagrams where these are relevant.

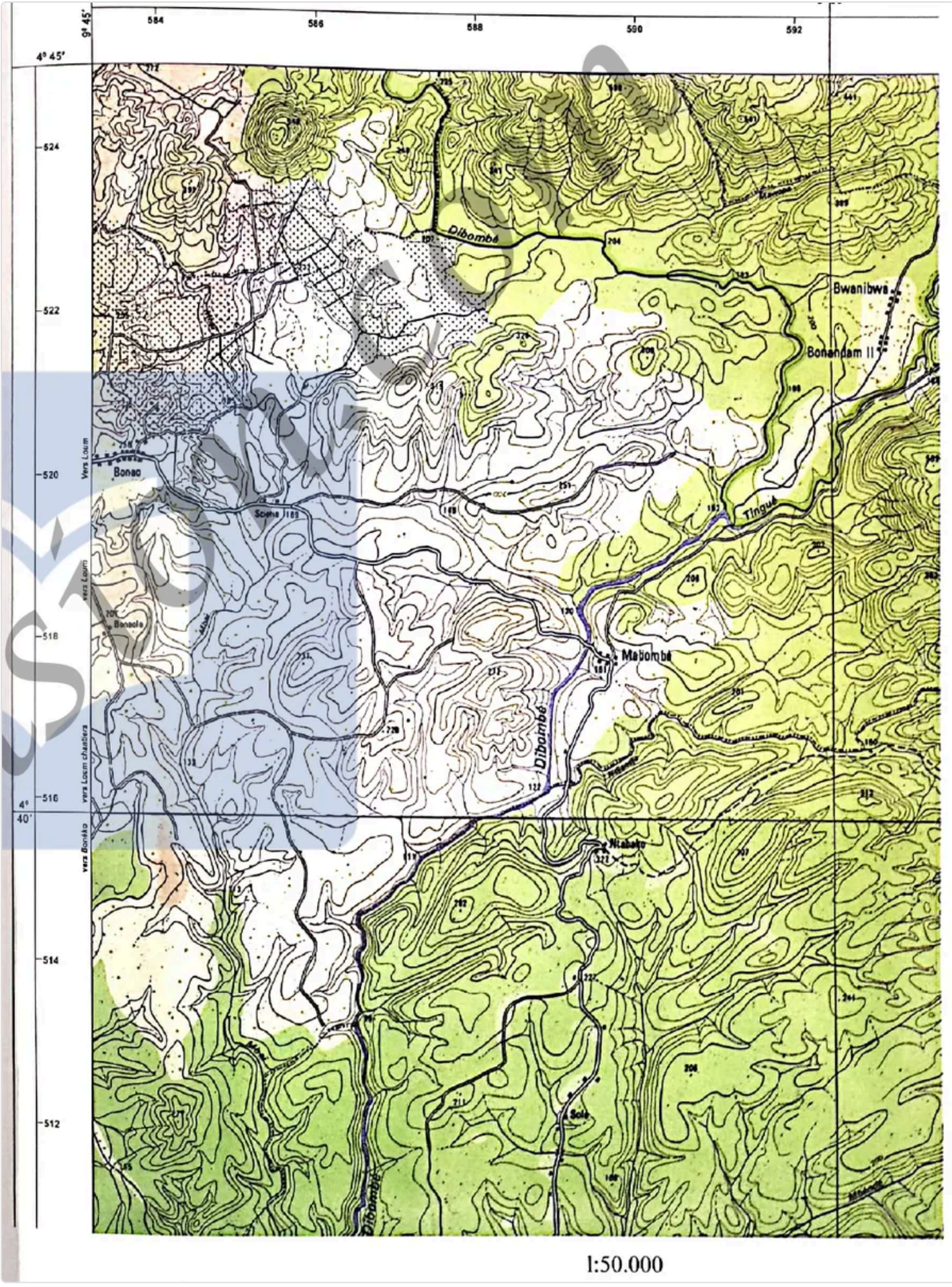
You are reminded of the necessity for good English and orderly presentation in your answers.

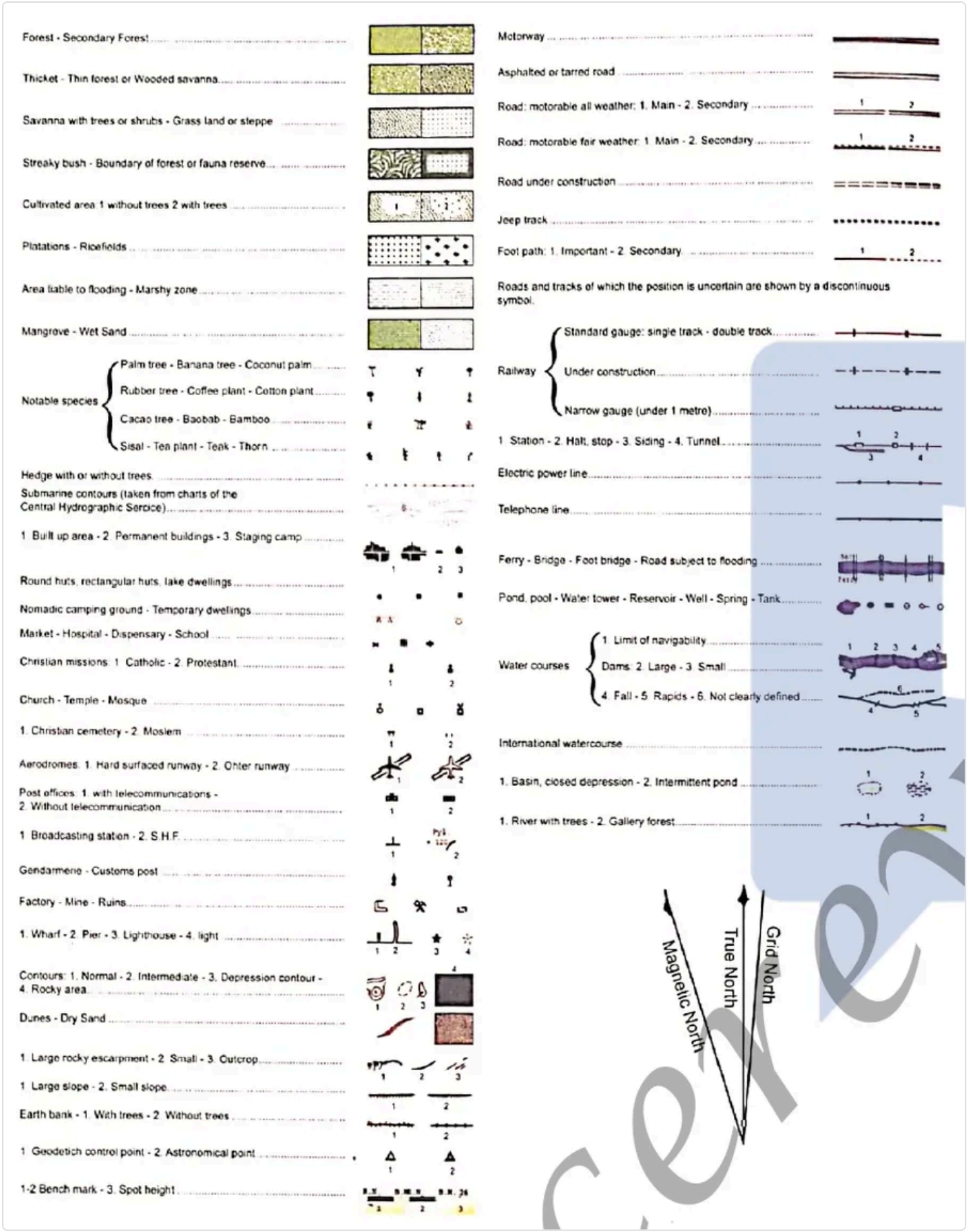
Calculators are allowed.

Materials required: Topographic maps, Graph Paper and two outline maps of Cameroon.

SECTION A (COMPULSORY) - MAP WORK

Study the topographic map extract of **Mabombe at 1:50.000** provided and answer the following questions.


9° 45'
584
586
588
590
592
4° 45'
524
522
520
518
516
514
512
4° 40'
Dibombé
Bwanibwa
Bonandam II
Bonao
Vers Loum
Tingub
Bonaole
Mabombé
Dibombé
Ntabako
Sole
1:50.000



[The scanned extract and its key above are partial and of low resolution; the full sheet was supplied on the day. Answers must be checked against the original sheet.]

a. a (i) Draw a sketch of the network of the stream that meets the River Dibombe south of $4^{\circ}40'$ at spot height 111 and rank the stream using Strahler's method. **(6 marks)**

(ii) Establish a frequency table of the stream order from the sketch drawn above. **(3 marks)**

(iii) Calculate the average Bifurcation Ratio for the stream pattern and interpret your result. **(5 marks)**

b. Briefly describe the distribution of vegetation on the whole map area. **(4 marks)**

c. Identify the patterns and describe the distribution of settlements on the whole map area. **(5 marks)**

d. d (i) Account for the location of the plantation in the north western section of the map. **(4 marks)**

(ii) Using evidence from the map, state THREE other economic activities performed in the area other than plantation farming. **(6 marks)**

SECTION B - FIELD WORK AND TECHNIQUES

1. You have carried out an investigative field work in one of the following topics:

2. River characteristics

3. Weather observations
4. Vegetation characteristics
5. Functional segregation
6. Shopping habits
7. Modern Factory or cottage (local) industry

Choose **one** of the topics and answer the following questions:

a) Using a sketch map, locate the area of your field investigation. **(4 marks)**

b) State your objective and the hypothesis that guided your investigation of the topic. **(2 marks)**

c) What tools did you use? **(3 marks)**

d) Describe how your data was collected. **(5 marks)**

e) How was the data analyzed and presented? **(6 marks)**

f) What were your findings and conclusion? **(3 marks)**

SECTION B - FIELD WORK AND TECHNIQUES (continued)

1. Study the table below showing the slope angle and soil depth on a farm in Wotutu village (Buea).

Sites	Slope angle (cm)	Soil depth (cm)	Rank slope	Rank soil depth	Differences (d)	Difference squared (d^2)
1	6	58	8	-	5	25
2	12	40	7	4	3	9
3	13	20	-	-	-3	-
4	14	18	5	10	-	25

Sites	Slope angle (cm)	Soil depth (cm)	Rank slope	Rank soil depth	Differences (d)	Difference squared (d^2)
5	5	60	9	-	7.5	56.25
6	23	25	-	7.5	-5.5	-
7	19	30	4	5	-	1
8	27	25	1	7.5	-6.5	42.25
9	20	29	-	6	-	-
10	3	60	10	-	8.5	72.25
						$\sum d^2 =$

(0.5 mk x 14 = 7 marks)

a) (i) Complete the table above by inserting the missing figures. **(7 marks)**

(ii) State the null hypothesis (Ho) tested on the table above. **(2 marks)**

b) (i) Calculate the correlation between slope angle and soil depth using the Spearman's Rank Correlation Coefficient (r_s) formula stated as:

$$r_s = 1 - \frac{6 \sum d^2}{n^3 - n}$$

where r_s is the Spearman correlation coefficient, d^2 is the square difference in rank, and n is the number of sites. **(4 marks)**

Study the levels of significance below.

N	Significance level, 95% confidence level	Significance level, 99% confidence level
4	1.00	-
5	0.90	1.00
6	0.83	0.94
7	0.71	0.89
8	0.64	0.83
9	0.60	0.78
10	0.56	0.75
12	0.51	0.71
14	0.46	0.65
16	0.43	0.60

(ii) State the statistical significance of the result in (b) (i). **(3 marks)**

c) (i) Describe and account for the relationship between slope angle and soil depth as portrayed on the table above. **(5 marks)**

(ii) State at least **two** importance of the Spearman's Rank Correlation Coefficient. **(2 marks)**

(Total = 23 marks)

SECTION C - CAMEROON

1. a) Explain how the movement of air masses causes the seasons in Cameroon? **(9 marks)**

b) (i) Briefly describe the spatial distribution and characteristics of plantation farming systems in Cameroon. **(8 marks)**

(ii) Describe any **TWO** strategies used by the government to tackle physical problems faced by peasant farmers in the Northern Lowlands. **(6 marks)**

Exercise 5 (23 marks)

1. a) Using the outline map of Cameroon provided, locate and briefly describe the main vegetation types in Cameroon. **(7 marks)**

[outline map of Cameroon supplied on the day, not reproduced here]

b) i) Identify major industries and their sources of raw materials in the Western Industrial Triangle. **(10 marks)**

ii) What efforts have been put in place to boost the industrial sector of Cameroon? **(6 marks)**

Exercise 6 (23 marks)

1. a) Consider the Coastal Lowlands of Cameroon and briefly describe its relief and drainage characteristics. **(8 marks)**

b) Examine the consequences of rural-urban migrations in Cameroon on the rural areas and communities. **(9 marks)**

c) Discuss at least **three** measures which have been undertaken to minimize such movements. **(6 marks)**

(Total=23 marks)

Exercise 7 (23 marks)

a. 7 a) Examine the distribution of water resources in Cameroon. **(7 marks)**

b) (i) Locate and describe the spatial pattern of rail way transport network in Cameroon. **(8 marks)**

(ii) Account for its present declining state. **(8 marks)**

(Total = 23 marks)

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

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